King Faisal University
College of Computer Sciences and Information Technology
Computer Science Department

[CS-210]: [Programming Methodologies]
Section: []

Lab [02]: [Variables, Data Types, Operators, and Output]

Date: []

Answer Sheet

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Computer Number:

Part 1 — Guided Debugging & Code Analysis

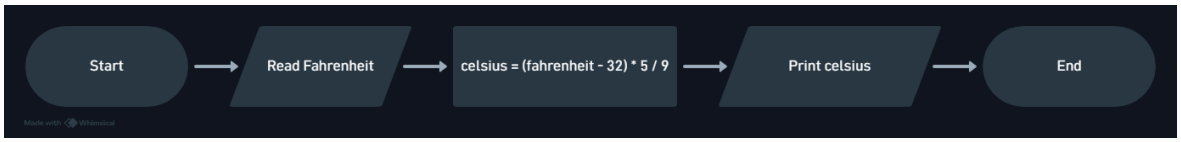
Error Identification & Correction Table

Fill in the table below by identifying the errors present in the code. One row is dedicated to each error.

Line #	Code Snippet with Error	Error Description (Student Fill)	Error Type (Syntax / Logical)	Corrected Code (Student Fill)
5	fahrenheit;	**variable not initialized**	**Syntax**	**double fahrenheit;**
6	**double celsius**	**no semicolon**	**Syntax**	**double celsius;**
7	**cout >>**	**wrong operator (>>)**	**Syntax**	**cout <<**
10	**celsius = 5 / 9 * (fahrenheit - 32);**	**integer division while float division needed**	**Logical**	**celsius = (float)5 / 9 * (fahrenheit - 32);**
11	**… << Celsius << …**	**Wrong identifier spelling**	**Syntax**	**… << celsius << …**

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



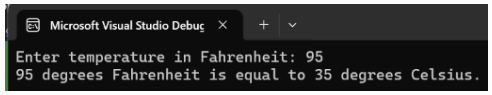
Corrected Source Code

Paste your complete, working, and corrected C++ source code in the box below.

```
//Temperature Converter (Fahrenheit to Celsius)
#include <iostream>
using namespace std;
int main() {
        double fahrenheit;
        double celsius;
        cout << "Enter temperature in Fahrenheit: ";
        cin >> fahrenheit;
        // Convert Fahrenheit to Celsius
        celsius = (float)5 / 9 * (fahrenheit - 32);
        cout << fahrenheit << " degrees Fahrenheit is equal to " << celsius << " degrees Celsius." << endl;
        return 0;
}
```

Execution Screenshot

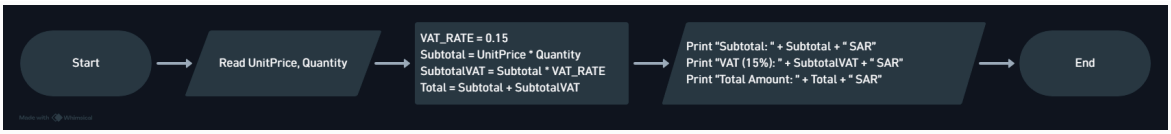
Paste a screenshot of your working program output here showing a successful temperature conversion.



Part 2 — Independent Program Implementation

Flowchart Diagram

Draw the flowchart of the given problem in the box below.



C++ Source Code:

Paste your complete C++ source code for the VAT calculator in the box below.

```cpp
#include <iostream>
using namespace std;

int main()
{
	const float VAT_RATE = 0.15;
	float UnitPrice;
	int Quantity;

	cout << "Enter unit price: ";
	cin >> UnitPrice;

	cout << "Enter quantity: ";
	cin >> Quantity;

	float Subtotal, SubtotalVAT, Total;

	Subtotal = UnitPrice * Quantity;
	SubtotalVAT = Subtotal * VAT_RATE;
	Total = Subtotal + SubtotalVAT;

	cout << endl; // newline to see the result better.
	cout << "Subtotal: " << Subtotal << " SAR\n";
	cout << "VAT (15%): " << SubtotalVAT << " SAR\n";
	cout << "Total Amount: " << Total << " SAR\n";

	return 0;
}
```

Execution Screenshot:

Paste a screenshot of your compiled VAT calculator running successfully with test data.

```
Enter unit price: 100
Enter quantity: 3

Subtotal: 300 SAR
VAT (15%): 45 SAR
Total Amount: 345 SAR
```